

SmartCare v0.1 - Initial Engineering Brief and AI Activity Card

A2 Case Study Stage 1 student resource

SmartCare scenario

SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The client says: 'We need software to help manage patients, practitioners and appointments.' This is not yet a complete specification.

Initial Engineering Brief

1. Problem summary

SmartCare Community Clinic relies on manual paper records and spreadsheets for scheduling and patient tracking, creating operational friction that limits efficiency, accuracy, and accessibility.

2. Initial stakeholders

Stakeholder	Possible need
Clinic Patients	Easy online appointment booking, clear schedule visibility, minimal wait times, and confidence that their personal health information remains secure and private.
Practice Manager / Administrative Staff	Fast patient search reduced manual double-entry, a reliable calendar to prevent double-booking, and a smooth transition from paper records with minimal operational downtime.
General Practitioners / Clinicians	Quick access to patient treatment histories before consultations, an intuitive daily schedule view, and fast, templated note-taking that doesn't distract from patient care.
Clinic Owner / Board	A cost-effective deployment, minimal software maintenance overhead, compliance with healthcare privacy regulations, and improved efficiency.

3. Initial features

Feature	Confirmed or provisional?	Why?
Appointment Scheduling & Calendar Management	Confirmed	Directly solves the main bottleneck by replacing paper-based systems and spreadsheets with a centralized digital calendar.
Digital Patient Profiles & Central Search	Confirmed	Moving away from physical filing cabinets requires a digital database.
Digitized Clinical Consultation Notes & Templates	Provisional	Clinicians need a structured way to record treatment plans and

		consultation notes without using paper.
SMS / Email Appointment Reminders	Provisional	Automated reminders significantly reduce patients forgetting their appointments.

4. Questions for the client

1. How many active patient records and upcoming appointments currently exist on paper or spreadsheets, and in what condition is the data for migration?
2. What device hardware (e.g., desktops, tablets) is currently available in the clinic, and what is the general digital-literacy level of the staff?
3. Are there specific regional healthcare privacy regulations we must comply with regarding where patient records are hosted?
4. Is there an allocated budget for recurring operational costs (Cloud hosting, SMS reminder service etc.)?
5. Would you prefer a phased rollout, or a 'one-shot' single transition, and how do we measure 'success'?

5. What we do not yet know

1. Current Data Quality and File Formats
2. Regulatory and Compliance Framework
3. Clinic Hardware Infrastructure and Technical Literacy

AI Activity Card - Ask, Check, Explain

Before AI

FILE: lab/smartcare_v01.py

What do I think the code does? What problems can I already identify?

Program is a basic appointment booking system. It lets users store appointments in a list, add new appointments through a menu and view all existing appointments.

AI request

Act as a tutor. Explain this code and identify potential problems. Do not provide a complete replacement. Ask me questions that help me reason about the solution.

CoPilot Output:

High-Level Overview

The program implements a very simple appointment booking system:

- Creates a global list called appointments.
- Stores each appointment as a dictionary.
- Provides a book() function to add appointments.
- Provides a display() function to show appointments.
- Uses a menu loop in main() to allow user interaction.
- Pre-loads two sample appointments when the program starts

Potential Problems

- Global Variable Dependency
- Weak Data Representation
- No Validation
- Date and Time Stored as Text
- No Duplicate Checking
- Mixed Responsibilities
- Return Value Not Used
- Automatic Sample Data
- No Persistence
- main() Runs Immediately

Questions

What real-world concepts exist in the problem domain?

- Patient?
- Practitioner?

- Appointment?

Which of those concepts have their own data and behaviour?

Evaluate

Suggestion	Useful	Unclear	Incorrect	Out of scope
Replace Appointment Dictionaries with an Appointment Class	Accept			
Validate User Input Before Storing It	Accept			
Avoid Direct Use of Global Variables	Accept			
Add Business Rules for Appointment Conflicts	Accept			

Decide

For each significant suggestion: Accept / Modify / Reject / Keep unverified.

See above

Verify

- Run the code
- Test normal input
- Test unusual input
- Compare with requirements
- Ask tutor/peer
- Check documentation

Explain

Can I explain the final code without reading the AI response? What do I still need to understand?

Yes, I can explain the final code, I still need to understand user validation and OOP but that will be left to later lectures.